

AMENDMENT NO. 3 DECEMBER 1988

TO

IS : 9974 (Part 1) - 1981 SPECIFICATION FOR HIGH PRESSURE SODIUM VAPOUR LAMPS

PART 1 GENERAL REQUIREMENTS AND TESTS

(Page 4, clause 2.7) — Insert the following new clauses after 2.7

2.8 Test Quantities

2.8.1 Inspection Test Quantity (ITQ) — The number of lamps selected for the purpose of determining the acceptability of a batch as to marking, mechanical and physical requirements.

2.8.2 Rating Test Quantity (RTQ) — The number of lamps selected for the purpose of determining the acceptability of a batch as to initial readings.

2.8.3 Life Test Quantity (LTQ) — The number of lamps selected for the purpose of determining the acceptability of a batch as to life performance.'

(Page 5, clause 4) — Substitute the following for the existing clause:

4. MECHANICAL AND PHYSICAL REQUIREMENTS

4.1 Lamp Dimensions — The lamp dimensions shall comply with the requirements given in the relevant lamp data sheet in Part 2 of this standard.

4.2 Glass Bulb — Glass bulbs of the lamps shall be free from defects detrimental to service.

4.3 Caps

4.3.1 The dimensions of the E40 caps on the finished lamp shall comply with the requirements given in IS : 9900 (Part 3)-1981*. The dimensions of the caps shall be checked with the 'Go' and 'No-Go' gauges of lamp caps as specified in IS : 9900 (Part 4)-1981†.

4.3.2 The shells of the caps may be made of any suitable material provided the caps are ordinarily resistant to atmospheric corrosion.

4.3.3 The cap shall be so constructed and attached to the bulb that it withstands the torsion test specified in Appendix A of IS : 9900 (Part 1)-1981‡.

4.3.4 Insulation Resistance — The insulation resistance between the shell of the bayonet cap and the contact shall be not less than 50 megohms. Measurement shall be made one minute after the application of a dc voltage of 500 V. Prior to test, the external surface of the insulation of

the cap shall be wiped with a dry cloth. In case the insulation resistance, after first measurement, is found to be insufficient, the lamps shall be stored in a warm, dry atmosphere for a week and then retested for insulation.'

(Page 5, clause 5.1) — Insert the following new clause after 5.1:

'5.1.1 The lamps may also be marked with the Standard Mark.

NOTE — The use of the Standard Mark is governed by the provisions of the Bureau of Indian Standards Act, 1986 and the Rules and Regulations made thereunder. The Standard Mark on products covered by an Indian Standard conveys the assurance that they have been produced to comply with the requirements of that standard under a well-defined system of inspection, testing and quality control which is devised and supervised by BIS and operated by the producer. Standard marked products are also continuously checked by BIS for conformity to that standard as a further safeguard. Details of conditions under which a licence for the use of Standard Mark may be granted to manufacturers or producers, may be obtained from the Bureau of Indian Standards'

(Page 5, clause 6) — Substitute the following for the existing clause:

'6. SELECTION OF LAMPS FOR TESTS (SAMPLING)

6.1 Method of Selection — The inspection and rating test quantities specified in the following clauses shall be selected in a mutually agreed manner such as to ensure proper representation of the batch.

6.1.1 The selection of lamps for individual batches should be made as follows:

- a) *Up to and including 8 containers per batch* — out of every container an equal number of lamps (or as near to equal as possible) shall be selected at random in order to obtain the 8 lamps required.
- b) *Over 8 containers per batch* — out of 8 containers, evenly distributed over the whole batch, one lamp shall be selected at random from each container to obtain the 8 lamps required.

6.2 Inspection Test Quantity (ITQ) — Inspection test quantity shall consist of 8 lamps.

6.3 Rating Test Quantity (RTQ) — Rating test quantity consisting of 6 lamps, shall be selected at random from the lamps which have passed the inspection tests.

6.4 Life Test Quantity (LTQ) — From the lamps which have passed rating test, they shall be selected at random a LTQ of 4 lamps.

(Page 5, foot-notes) — Insert the following foot-note in the end:

[†]Specification for high pressure mercury vapour lamps: Part I Requirements and tests.

(Page 9, clause 9.3) — Add the following new clause after 9.3:

10. REQUIREMENTS AND CONDITIONS OF TEST FOR LIFE PERFORMANCE

10.1 After 5 000 hours operation, including the ageing period of 100 hours, the lumen maintenance shall not be less than the value indicated below:

- | | |
|--|-------------|
| a) For lamps up to and including 400 W | 80 percent. |
|--|-------------|

10.2 The life test shall be carried out on an ac supply with a declared frequency of 50 Hz which shall be the rated frequency of the ballast. The test voltage shall be equal to the rated voltage of the ballast ± 1 percent.

For life testing, commercial ballasts with electrical parameters not differing by more than 3 percent from the reference ballast can be used.

The life test shall be made at an ambient temperature between 15°C and 50°C.

The momentary fluctuations of the test voltage and frequency during the test shall not exceed ± 2 percent in each case.

10.2.1 *Switching 'on' and switching 'off' during life test* — Lamps on life test shall be switched 'off' after every $7\frac{1}{2}$ hours of burning, each 'off' period consisting of $\frac{1}{2}$ hour duration. The 'off' period shall not be included in calculating the number of burning hours. The life test, as far as possible, shall be done continuously.

11. CONDITIONS OF COMPLIANCE

11.1 General Conditions — A batch shall be considered as conforming to this standard if the requirements contained in 11.2, 11.3 and 11.4 are fulfilled. If the batch fails to satisfy the requirements of any of these clauses, it shall be deemed not to comply with the standard.

11.2 Mechanical, Physical and Marking Requirements — A batch shall be considered to comply with the requirements of 4 and 5, if the number of lamps failing does not exceed qualifying limits given below:

Qualifying Limit

- | | |
|--|---|
| a) For any single requirement | 1 |
| b) For all requirements taken together | 2 |

11.3 Lamp Starting, Warm-up and Electrical Characteristics

11.3.1 A batch shall be considered to comply with the requirements of lamp starting, warm-up and electrical characteristics if the number of lamps failing does not exceed the qualifying limits given below:

	<i>Qualifying Limit</i>
a) For any single requirement	1
b) For all requirements taken together	2

11.4 Lumen Maintenance — A batch shall be considered to comply with the requirements of life if the total number of lamps having lives shorter than 5 000 hours, together with those failing to meet the requirements of lumen maintenance given in 10.1 does not exceed the qualifying limits given below:

<i>Life Test Quantity</i>	<i>Qualifying Limit</i>
4	1

(ETDC 23)